CHAPTER IV

METHODOLOGY

This chapter presents the methodology used in developing the Smart Water Monitoring Consumption System. Following the sequence of the supplied Chapter 4 reference, it discusses requirements and use cases, system design and user interfaces, the entity relationship diagram, the data dictionary, development and testing, the implementation plan, encountered challenges, and implementation results. All technical statements are based on the implemented Laravel application, ESP32 firmware, database migrations, routes, services, views, and automated tests of the Smart Water project.

4.1 Requirements Analysis and Requirements Documentation

Requirements analysis identified three principal actors: the municipal administrator, the household account holder, and the commissioned smart meter. The administrator manages household records, municipal meters, water-rate configuration, billing statements, leak incidents, notifications, analytics, reports, and audit information. The household user reviews live and historical consumption, estimated billing, leak status, notifications, and account settings. The smart meter measures flow-sensor pulses and submits authenticated interval readings to the server.

4.1.1 Functional Requirements

ID	Requirement	Verified Project Behavior
FR-01	User and access management	Administrator creates and manages household accounts; public registration is restricted; account status governs access.
FR-02	Municipal device lifecycle	Register, assign, activate, rotate credentials, reassign, suspend, and decommission meters.

ID	Requirement	Verified Project Behavior
FR-03	Telemetry ingestion	Accept authenticated interval readings containing a stable reading identifier, flow rate, liters used, meter state, and device diagnostics.
FR-04	Consumption monitoring	Display live readings, daily/weekly/monthly history, summaries, comparisons, and availability states.
FR-05	Estimated billing	Apply the latest applicable water rate, fixed charge, subsidy, and billable consumption to estimates and monthly statements.
FR-06	Leak and threshold detection	Evaluate server-configured thresholds, notify affected users, and support an administrator resolution workflow.
FR-07	Notifications	Create, read, archive, retry, and remove household or administrative notices according to role and ownership.
FR-08	Reports and exports	Produce consumption, billing, leak, notification, and operational reports with filtering and export support.
FR-09	Audit and analytics	Record administrative/security activity and present household- or administrator-scoped analytics.
FR-10	Account maintenance	Allow authenticated users to update permitted profile fields and change passwords.

4.1.2 Non-Functional Requirements

Quality Attribute	System Response
Security	Laravel Sanctum tokens, role middleware, scoped device credentials, hashed one-time activation codes, token revocation, and audit redaction.
Reliability	Stable reading identifiers prevent double counting; unsent meter intervals are retained in ESP32 Preferences until acknowledged.
Performance	The meter uploads at three-second intervals and the web interface is designed to refresh live information at the same cadence.

Quality Attribute	System Response
Data integrity	Foreign keys, validation, unique identifiers, server-owned household/device associations, precision-preserving numeric fields, and historical charge snapshots.
Usability	Responsive administrator and household workspaces, accessible labels, keyboard-oriented controls, role identification, and light/dark theme support.
Maintainability	Controllers are grouped by feature; models represent domain entities; services isolate billing, analytics, consumption, and audit logic.
Portability	The web application supports SQLite or MySQL, while the field device targets ESP32-S3/M5Stack CoreS3 hardware.

4.1.3 Use-Case Diagram

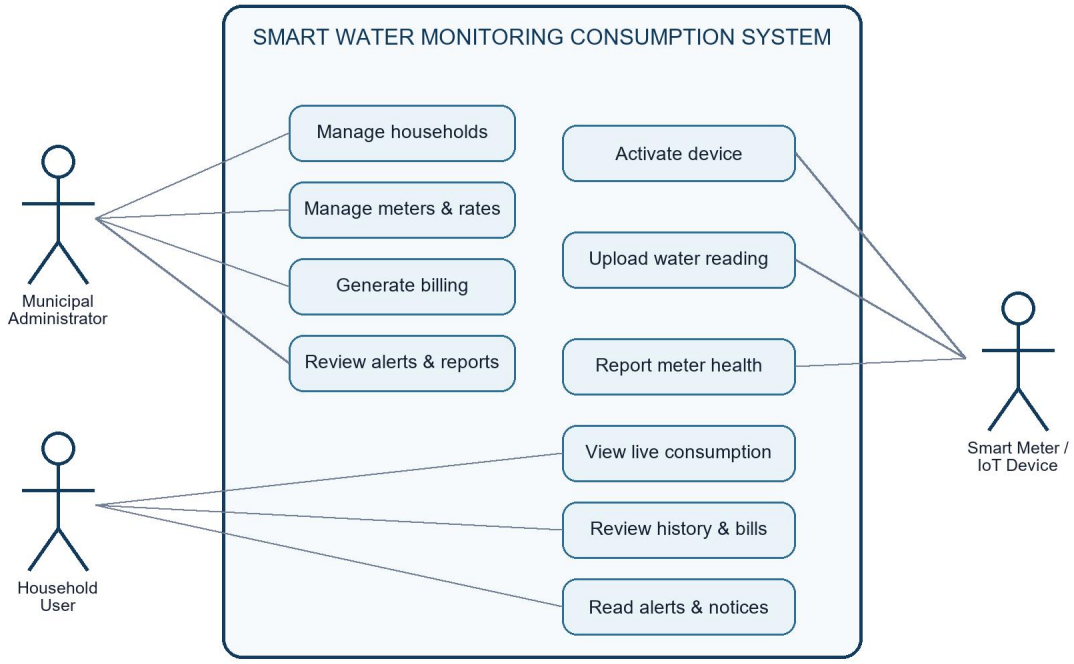


Figure 4.1. Use-case overview of the Smart Water Monitoring Consumption System.

4.1.4 Use-Case Descriptions

The following use-case descriptions adapt the detailed documentation format shown in the reference to the implemented Smart Water actors and workflows. Each specification identifies the responsible actor, scenario, trigger, normal sequence, supporting path, and exceptional conditions.

4.1.4.1 Manage Households

Use Case Name: Manage Households	ID: UC-01	Importance Level: High
Primary Actor: Municipal Administrator	Use Case Type: Essential	
Scenario: The administrator wants to create, review, update, or change the operational status of a household account.		
Brief Description: Maintains household identity, service address, contact information, account number, access status, and meter-assignment readiness.		
Trigger: The administrator selects Customer Management from the administrative workspace.		
Normal Flow of Events: 1. The administrator signs in and opens Customer Management. 2. The system displays searchable and filterable household records. 3. The administrator creates a new household or selects an existing record. 4. The administrator enters or updates the permitted household information. 5. The system validates required fields, unique account details, role, and status. 6. The system saves the household record and logs the administrative activity.		
Sub Flows: The administrator may suspend, deactivate, archive, reactivate, or unassign the household meter when authorized.		
Alternative/Exceptional Flows: Duplicate emails or account numbers, missing required fields, and unauthorized role changes are rejected with a validation response.		

Table 4.1.1 Manage Households

4.1.4.2 Manage Meters and Rates

Use Case Name: Manage Meters and Rates	ID: UC-02	Importance Level: High
Primary Actor: Municipal Administrator	Use Case Type: Essential	
Scenario: The administrator wants to manage municipal smart meters and configure the water rates and monitoring rules applied to households.		
Brief Description: Combines meter registration, assignment, maintenance, credential control, and effective-dated water-rate and alert configuration.		
Trigger: The administrator opens Device Management or Water Rate Management.		
Normal Flow of Events: 1. The administrator opens the required meter or rate module. 2. The system displays the current devices or active billing configurations. 3. For a meter, the administrator registers, assigns, reassigns, suspends, rotates credentials, or decommissions the asset. 4. For a rate, the administrator enters the water rate, fixed charge, subsidy, currency, effective date, and alert thresholds. 5. The system validates identifiers, household eligibility, numeric ranges, and effective dates. 6. The system saves the approved meter or rate configuration and records the activity.		
Sub Flows: The administrator may review meter health, calibration dates, activation state, last-seen status, or household-specific rate overrides.		
Alternative/Exceptional Flows: Duplicate assets, ineligible household assignments, invalid rates, negative charges, and unauthorized changes are rejected.		

Table 4.1.2 Manage Meters and Rates

4.1.4.3 Generate Billing

Use Case Name: Generate Billing	ID: UC-03	Importance Level: High
Primary Actor: Municipal Administrator	Use Case Type: Essential	
Scenario: The administrator wants to generate a monthly billing statement for an eligible household.		
Brief Description: Aggregates billable consumption, applies the applicable rate configuration, and preserves a historical charge snapshot.		
Trigger: The administrator chooses Generate Statement in Billing Management.		
Normal Flow of Events: 1. The administrator opens Billing Management. 2. The administrator selects the household and billing month. 3. The system validates the period, household, billable readings, and applicable rate. 4. The billing service totals liters and converts the result to cubic meters. 5. The service applies the water rate, fixed charge, subsidy, and opening balance. 6. The system stores the statement, charge breakdown, remaining balance, and payment status. 7. The administrator reviews the generated billing record.		
Sub Flows: Authorized staff may record payments and update the statement status through the implemented billing workflow.		
Alternative/Exceptional Flows: Duplicate billing months, missing billable readings, or unavailable rate configurations produce a specific response without creating a misleading statement.		

Table 4.1.3 Generate Billing

4.1.4.4 Review Alerts and Reports

Use Case Name: Review Alerts and Reports	ID: UC-04	Importance Level: High
Primary Actor: Municipal Administrator	Use Case Type: Essential	
Scenario: The administrator wants to review leak or consumption alerts and examine municipal operational reports.		
Brief Description: Supports alert investigation, incident follow-up, analytics, filtered reports, and supported data exports.		
Trigger: The administrator opens Leak Monitoring, Reports, Analytics, or the administrative notification view.		
Normal Flow of Events: 1. The administrator selects the alert or report module. 2. The administrator chooses the affected household, report type, date range, or filter. 3. The system validates authorization and input parameters. 4. The system retrieves the related readings, alerts, billing information, and operational summaries. 5. The administrator reviews the evidence and records follow-up notes when an incident is involved. 6. The administrator resolves, marks false, or reopens an alert, or exports a supported report.		
Sub Flows: The administrator may review notification delivery state, analytic summaries, and audit context related to an operational event.		
Alternative/Exceptional Flows: Invalid ranges, unsupported exports, missing incident identifiers, and unauthorized access return clear validation or access responses.		

Table 4.1.4 Review Alerts and Reports

4.1.4.5 View Live Consumption

Use Case Name: View Live Consumption	ID: UC-05	Importance Level: High
Primary Actor: Household User	Use Case Type: Essential	
Scenario: The household user wants to view the latest water consumption and meter state.		
Brief Description: Displays the latest owned reading, current flow, interval consumption, estimated billing state, leak state, and meter availability.		
Trigger: The household user opens the Dashboard or Live Monitoring page.		
Normal Flow of Events: 1. The household user signs in. 2. The user opens Dashboard or Live Monitoring. 3. The browser requests the latest household-scoped readings. 4. The server verifies ownership and retrieves the current meter and consumption state. 5. The interface displays current flow, liters used, estimated bill, leak status, meter connection, and recent activity. 6. The interface refreshes according to the configured near-real-time interval.		
Sub Flows: The user may filter recent readings by date or displayed row count.		
Alternative/Exceptional Flows: When no active meter or recent reading exists, the interface displays a specific unavailable state rather than misleading zero consumption.		

Table 4.1.5 View Live Consumption

4.1.4.6 Review History and Bills

Use Case Name: Review History and Bills	ID: UC-06	Importance Level: High
Primary Actor: Household User	Use Case Type: Essential	
Scenario: The household user wants to review past consumption and current or finalized billing information.		
Brief Description: Combines daily, weekly, monthly, or custom consumption history with estimated billing details and monthly statements.		
Trigger: The user opens Consumption History or Estimated Billing.		
Normal Flow of Events: 1. The user selects the consumption-history or billing page. 2. For history, the user selects a period or valid custom date range. 3. The server retrieves only the household's readings and computes totals, comparisons, and chart data. 4. For billing, the server selects the applicable rate and eligible consumption information. 5. The interface displays usage summaries, trends, the estimated charge breakdown, and finalized statement history. 6. The user reviews the selected history or billing record.		
Sub Flows: The household may switch among daily, weekly, and monthly views and inspect individual monthly statements.		
Alternative/Exceptional Flows: Invalid ranges are rejected. Missing readings or rate information remain unavailable rather than being interpreted as zero usage or zero charges.		

Table 4.1.6 Review History and Bills

4.1.4.7 Read Alerts and Notices

Use Case Name: Read Alerts and Notices	ID: UC-07	Importance Level: Medium
Primary Actor: Household User	Use Case Type: Essential	
Scenario: The household user wants to read leak, consumption, billing, meter, or account notifications.		
Brief Description: Presents household-scoped alerts and official notices while preserving the municipality's source and delivery records.		
Trigger: The user opens Notifications or selects a recent alert from the household workspace.		
Normal Flow of Events: 1. The household user signs in and opens Notifications. 2. The system retrieves notices owned by the household account. 3. The interface displays each notice's type, title, message, date, and read state. 4. The user opens a selected alert or notice. 5. The system marks the notice as read when applicable. 6. The user may remove the notice from the household view.		
Sub Flows: Leak-related notices may direct the user to the Leak Detection page for the latest meter signal and response status.		
Alternative/Exceptional Flows: A household cannot access another account's notices. User-side removal hides the notice without deleting the municipality's source record.		

Table 4.1.7 Read Alerts and Notices

4.1.4.8 Activate Device

Use Case Name: Activate Device	ID: UC-08	Importance Level: High
Primary Actor: Smart Meter / IoT Device	Use Case Type: Essential	
Scenario: A registered M5Stack smart meter wants to exchange its one-time code for a scoped device credential.		
Brief Description: Binds the physical hardware to the municipal device record and activates authenticated telemetry access.		
Trigger: The technician enters the administrator-issued activation code on the registered device.		
Normal Flow of Events: 1. The administrator has already registered the meter and issued a short-lived activation code. 2. The device submits its UUID, activation code, hardware identity, and firmware version. 3. The server verifies that the code is valid, unused, and unexpired. 4. The server checks device status and hardware-binding rules. 5. The server binds the hardware identity and marks the device active. 6. The server issues a scoped device credential. 7. The M5Stack stores the credential and becomes ready to upload telemetry.		
Sub Flows: An administrator may later rotate the credential or reissue activation after an authorized lifecycle change.		
Alternative/Exceptional Flows: Expired, reused, incorrect, or mismatched activation information is rejected and no device credential is issued.		

Table 4.1.8 Activate Device

4.1.4.9 Upload Water Reading

Use Case Name: Upload Water Reading	ID: UC-09	Importance Level: High
Primary Actor: Smart Meter / IoT Device	Use Case Type: Essential	
Scenario: An activated smart meter wants to submit a measured water-consumption interval.		
Brief Description: Authenticates the device, validates telemetry, stores the reading once, and returns authoritative totals and alert state.		
Trigger: The firmware reaches the configured upload interval with a pending measurement.		
Normal Flow of Events: 1. The sensor counts valid flow pulses during the sampling interval. 2. The firmware calculates flow rate and interval liters. 3. The device creates a stable reading identifier and stores the pending state. 4. The meter submits the reading with its scoped credential. 5. The server derives the device and household from the authenticated identity. 6. The server validates and stores the reading idempotently. 7. The server evaluates thresholds and returns the authoritative total and leak state. 8. After acknowledgement, the meter clears the pending interval and advances its sequence.		
Sub Flows: The payload may include device name, monitoring state, Wi-Fi strength, and connectivity information.		
Alternative/Exceptional Flows: When transmission fails, the pending reading remains available for retry. Duplicate reading identifiers do not double-count consumption.		

Table 4.1.9 Upload Water Reading

4.1.4.10 Report Meter Health

Use Case Name: Report Meter Health	ID: UC-10	Importance Level: Medium
Primary Actor: Smart Meter / IoT Device	Use Case Type: Supporting	
Scenario: The smart meter wants to report its connection, firmware, and operational health to the server.		
Brief Description: Updates municipal visibility of device connectivity, last-seen activity, telemetry status, firmware version, and relevant diagnostics.		
Trigger: The device activates, uploads telemetry, reconnects to the network, or reaches a health-reporting interval.		
Normal Flow of Events: 1. The M5Stack evaluates its Wi-Fi and server connection state. 2. The device prepares its hardware identity, firmware version, monitoring state, signal strength, and health metadata. 3. The device submits the health information using its scoped credential. 4. The server validates the authenticated device identity. 5. The server updates last-seen, last-telemetry, firmware, and operational health fields. 6. The administrator workspace displays the latest device status.		
Sub Flows: Health information may accompany a water-reading upload to reduce unnecessary network requests.		
Alternative/Exceptional Flows: Invalid credentials or mismatched hardware identity are rejected. During network loss, the device retains its state and reports again after reconnection.		

Table 4.1.10 Report Meter Health

4.2 System Design, Software, and Processes

The system uses an Internet of Things client-server design. The physical meter performs measurement and short-term persistence, while the Laravel server remains authoritative for identity, ownership, cumulative usage, billing rules, threshold evaluation, and record storage. The browser interfaces consume protected JSON endpoints and present role-specific workspaces.

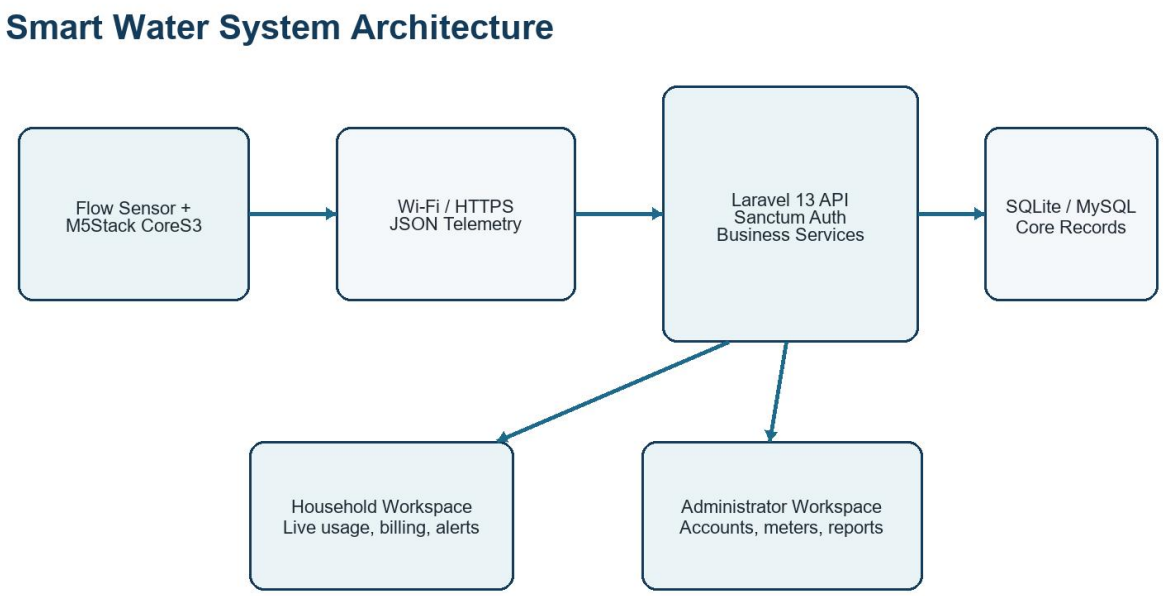


Figure 4.2. Smart Water Monitoring Consumption System architecture.

4.2.1 Telemetry and Billing Process

Commissioning begins in the administrator workspace. The device exchanges its UUID, expiring activation code, hardware ID, and firmware version for a scoped credential. Each reading includes a stable reading_id derived from the registered device identity and local sequence. If transmission fails, the pending interval remains stored in the device. When the API acknowledges the reading, the local sequence advances and the interval accumulator is cleared. The server validates ownership and computes authoritative totals before exposing information to users.

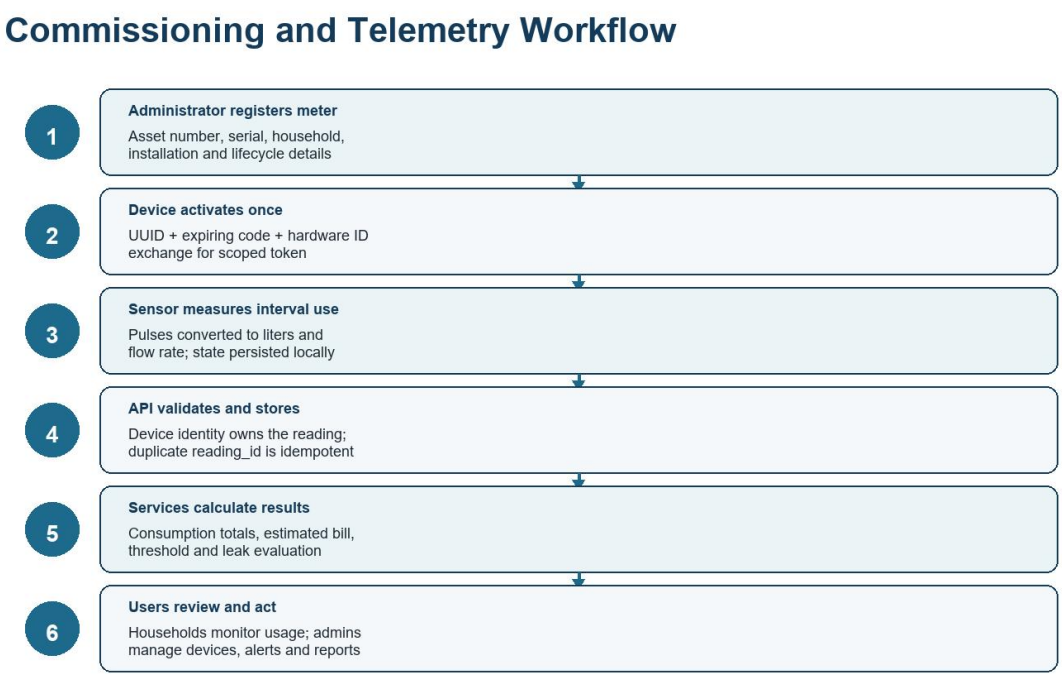


Figure 4.3. Device commissioning, telemetry, calculation, and user-response workflow.

4.2.2 Technology Used

Layer	Technology	Purpose
Backend	PHP 8.3+; Laravel Framework 13.8; Laravel Sanctum 4.0	Routing, validation, authentication, API controllers, services, reports, and persistence.
Frontend	Blade, JavaScript, Tailwind CSS 4, Vite 8	Responsive household and administrator workspaces, charts, filters, forms, and live refresh.
Database	SQLite or MySQL	Users, meters, readings, rates, billing statements, notifications, leak alerts, audit logs, sessions, and tokens.
Embedded device	ESP32-S3 / M5Stack CoreS3; M5Unified; Wi-Fi; HTTPClient; Preferences	Pulse measurement, device display, activation, reliable telemetry, and local pending-state storage.
Reporting	Dompdf 3.1.2 and CSV export endpoints	Printable operational reports and downloadable tabular records.
Testing	PHPUnit 12.5.12 through Laravel test runner	Feature verification for access control, telemetry, billing, alerts, analytics, devices, and interfaces.

4.2.3 Output and User Interface Design

The output and user interface design separates household monitoring from municipal administration. Household screens prioritize current flow, consumption history, billing estimates, leak status, notifications, and account settings. Administrator screens prioritize customers, devices, rates, statements, consumption, leak incidents, reports, notifications, analytics, and audit information. The responsive Blade layouts maintain consistent navigation while preserving role-based access boundaries.

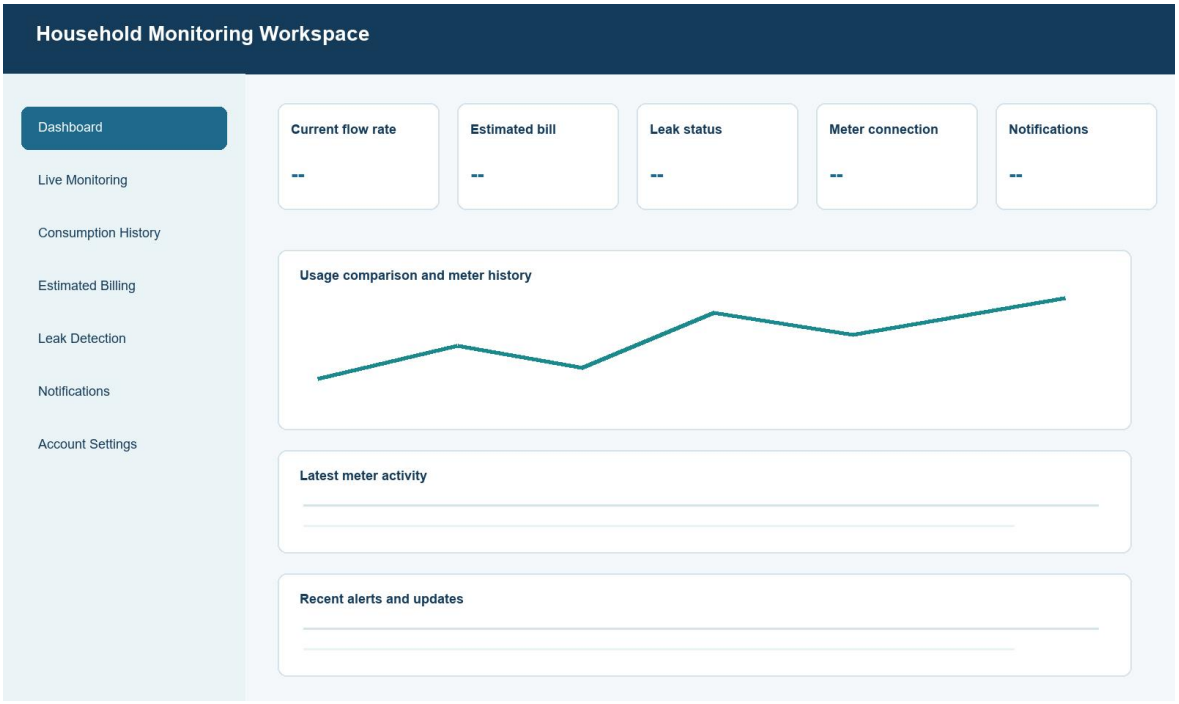


Figure 4.4. Household workspace interface map.

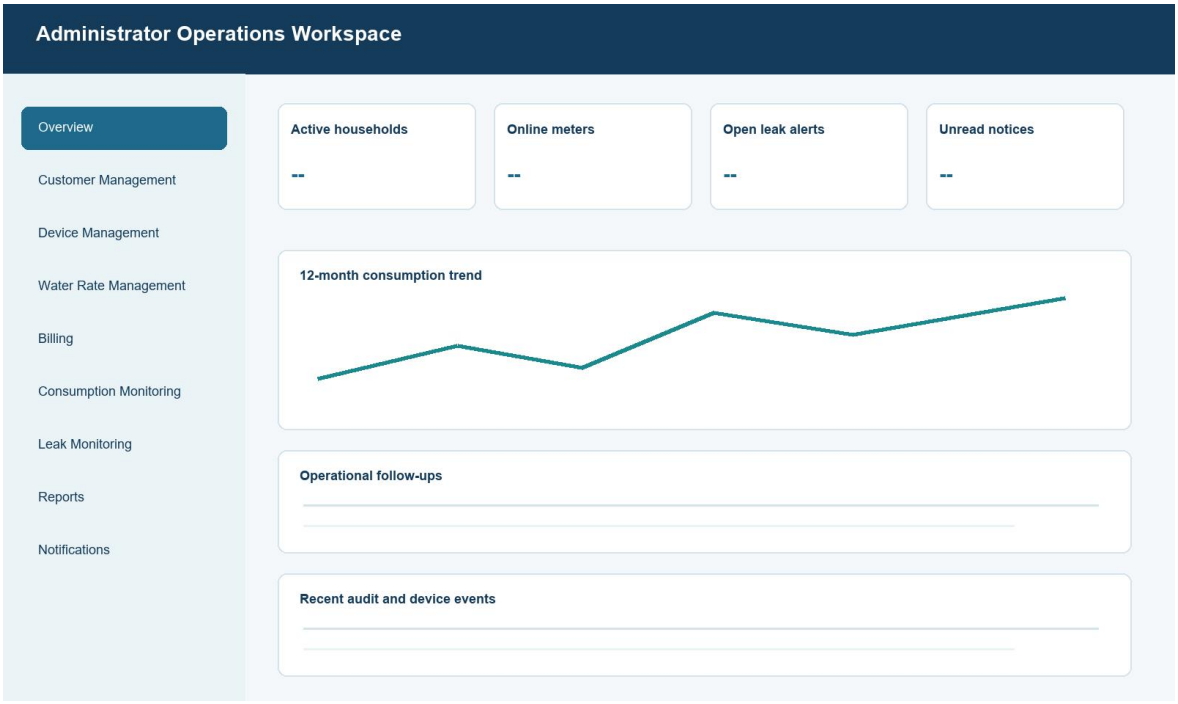


Figure 4.5. Administrator workspace interface map.

User Interface	Primary Outputs	Design Purpose
Household dashboard	Current flow, usage, estimated bill, meter status, leak state, and recent notices	Present essential account information at a glance.
Live monitoring and history	Recent readings, charts, period totals, summaries, and availability states	Support near-real-time review and time-based consumption analysis.
Billing and alerts	Rate breakdown, statements, leak status, thresholds, and notifications	Make charges and exceptional conditions understandable and actionable.
Administrator workspace	Household, meter, rate, billing, consumption, alert, report, and audit records	Centralize municipal operations while maintaining traceability.

4.3 Entity Relationship Diagram

The principal database relationships connect household users to their assigned devices, readings, statements, notifications, and leak incidents. A device may be temporarily unassigned during municipal inventory or maintenance, but an uploaded reading is tied to both the authenticated device and its current household. Billing configuration can be municipal or user-specific, and finalized statements preserve the charge values applied at generation time.

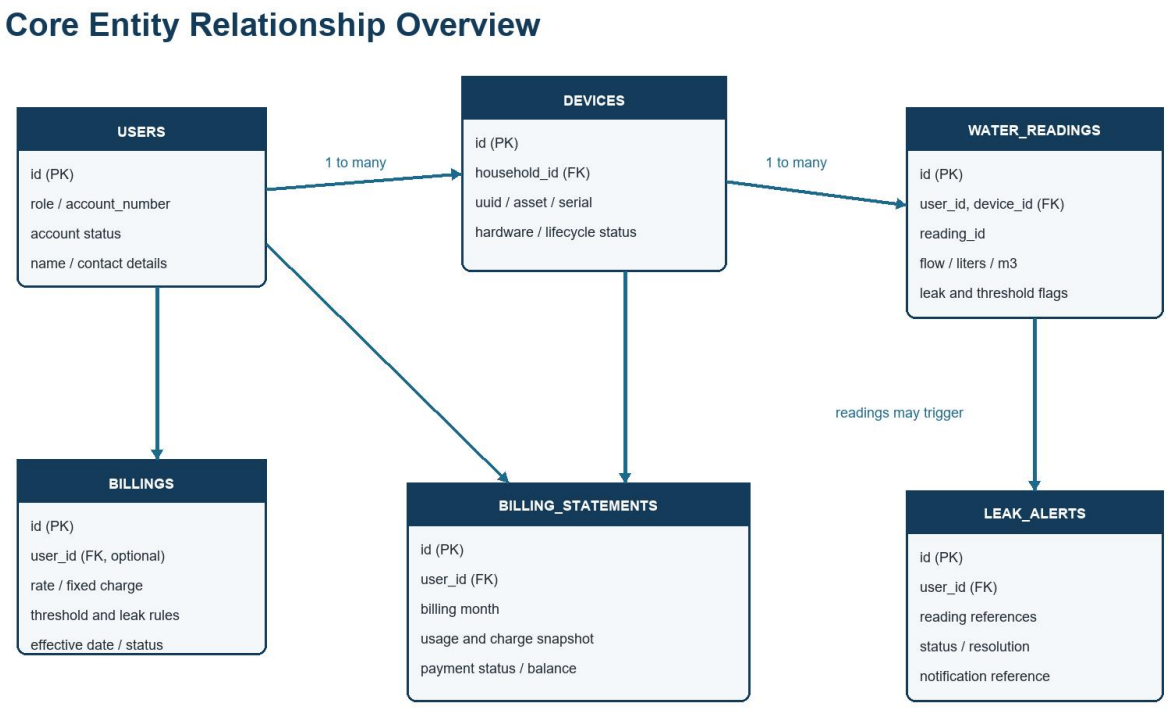


Figure 4.6. Core entity relationship overview (operational fields summarized).

4.4 Data Dictionary

The following data dictionary summarizes the operational tables most directly involved in monitoring, billing, alerting, and administration. Framework-support tables for cache, jobs, sessions, password resets, and personal access tokens remain part of the implementation but are omitted from the main discussion for clarity.

4.4.1 Users Table

The users table stores users data used by the implemented Smart Water application.

Field Name	Data Type	Length	Description
id	BIGINT	20	Primary key of the user account.
name	VARCHAR	255	Display name of the administrator, household, or device account.
email	VARCHAR	255	Unique login or account email address.
role	VARCHAR	255	Authorization role: administrator, household user, or device.
account_number	VARCHAR	255	Standardized municipal account identifier.
address	VARCHAR	255	Required service address for household accounts.
phone	VARCHAR	255	Optional contact number.
is_active	BOOLEAN	1	Indicates whether the account can remain operational.
email_verified_at	DATETIME	-	Date and time the email was verified.
password	VARCHAR	255	Hashed interactive-account password.
remember_token	VARCHAR	100	Framework token used for remembered sessions.
created_by_admin	BOOLEAN	1	Indicates municipal administrator provisioning.
assigned_user_id	BIGINT	20	Household linked to a legacy or supporting device account.
created_by	BIGINT	20	Administrator who created the account.
approved_at	DATETIME	-	Date and time the household account was approved.
account_status	VARCHAR	255	Lifecycle state such as active, suspended, archived, or deactivated.
created_at	DATETIME	-	Record creation timestamp.
updated_at	DATETIME	-	Record update timestamp.

Table 4.4.1 Database Table of Users

4.4.2 Devices Table

The devices table stores devices data used by the implemented Smart Water application.

Field Name	Data Type	Length	Description
id	BIGINT	20	Primary key of the municipal device record.
uuid	VARCHAR	255	Unique server-issued device identifier.
asset_number	VARCHAR	255	Municipal equipment inventory number.
serial_number	VARCHAR	255	Unique manufacturer or field serial number.
name	VARCHAR	255	Human-readable meter name.
household_id	BIGINT	20	Currently assigned household; nullable while in inventory.
auth_user_id	BIGINT	20	Limited authentication identity associated with the device.
hardware_id	VARCHAR	255	Bound ESP32 hardware identity.
firmware_version	VARCHAR	255	Firmware version reported during activation or telemetry.
status	VARCHAR	255	Lifecycle state such as pending, active, suspended, or decommissioned.
maintenance_status	VARCHAR	255	Operational or maintenance condition.
installation_address	VARCHAR	255	Physical service location of the meter.
installed_at	DATETIME	-	Installation date and time.
last_seen_at	DATETIME	-	Latest successful device contact.
last_telemetry_at	DATETIME	-	Latest accepted telemetry timestamp.
calibrated_at	DATETIME	-	Most recent calibration date.
calibration_due_at	DATETIME	-	Scheduled calibration due date.
token_rotated_at	DATETIME	-	Latest credential rotation timestamp.
revoked_at	DATETIME	-	Credential or device revocation timestamp.
activation_code_hash	VARCHAR	255	Hash of the one-time activation code.
activation_expires_at	DATETIME	-	Expiration of the activation code.
activated_at	DATETIME	-	Successful activation timestamp.
notes	TEXT	-	Administrative installation or maintenance notes.
created_at	DATETIME	-	Record creation timestamp.
updated_at	DATETIME	-	Record update timestamp.

Table 4.4.2 Database Table of Devices

4.4.3 Water Readings Table

The water_readings table stores water readings data used by the implemented Smart Water application.

Field Name	Data Type	Length	Description
id	BIGINT	20	Primary key of the telemetry record.
user_id	BIGINT	20	Household owner of the reading.
device_id	BIGINT	20	Authenticated meter that submitted the reading.
reading_id	VARCHAR	255	Stable idempotency identifier generated by the device.
flow_rate	DECIMAL	18,6	Measured flow rate in liters per minute.
liters_used	DECIMAL	18,6	Water volume represented by the interval.
cubic_meter	DECIMAL	18,6	Consumption converted to cubic meters.
estimated_bill	DECIMAL	12,2	Estimated charge stored with the reading.
rate_per_cubic_meter	DECIMAL	12,4	Applicable water rate used for the estimate.
subsidy_deduction	DECIMAL	12,2	Subsidy applied to the estimate.
threshold_liters	DECIMAL	18,6	Consumption threshold applicable at processing time.
is_threshold_exceeded	BOOLEAN	1	Indicates consumption threshold exceedance.
is_possible_leak	BOOLEAN	1	Indicates possible leakage under server rules.
leak_detection_status	VARCHAR	255	Detailed leak evaluation state.
water_limit_liters	DECIMAL	18,6	Configured household water limit snapshot.
monitoring_status	VARCHAR	255	Device monitoring state reported with telemetry.
wifi_signal_strength	INTEGER	11	Reported Wi-Fi signal strength.
device_connection_status	VARCHAR	255	Meter-side connection state.
server_connection_status	VARCHAR	255	Latest server acknowledgement state.
device_name	VARCHAR	255	Reported or display device name.
device_payload	JSON/TEXT	-	Additional validated device diagnostics.
recorded_at	DATETIME	-	Measurement timestamp supplied or normalized by the server.
created_at	DATETIME	-	Record creation timestamp.
updated_at	DATETIME	-	Record update timestamp.

Table 4.4.3 Database Table of Water Readings

4.4.4 Billing Configurations Table

The billings table stores billing configurations data used by the implemented Smart Water application.

Field Name	Data Type	Length	Description
id	BIGINT	20	Primary key of the billing configuration.
user_id	BIGINT	20	Household override owner; null indicates municipal default.
water_rate	DECIMAL	12,4	Charge per cubic meter.
fixed_monthly_charge	DECIMAL	12,2	Fixed recurring monthly charge.
subsidy_deduction	DECIMAL	12,2	Configured subsidy deduction.
currency	VARCHAR	3	Three-letter billing currency, normally PHP.
effective_date	DATE	-	Date on which the configuration becomes applicable.
threshold_liters	DECIMAL	18,6	Consumption alert threshold.
consumption_alert_enabled	BOOLEAN	1	Enables consumption threshold notifications.
consumption_alert_unit	VARCHAR	255	Unit used by the consumption threshold.
consumption_alert_behavior	VARCHAR	255	Audience or handling rule for consumption notices.
leak_alert_enabled	BOOLEAN	1	Enables leak alert evaluation.
leak_flow_rate_threshold	DECIMAL	18,6	Flow-rate threshold used for possible leak detection.
leak_continuous_duration_minutes	INTEGER	11	Required continuous duration before leak handling.
leak_alert_behavior	VARCHAR	255	Audience or handling rule for leak notices.
quiet_hours_start	INTEGER	2	Starting hour of the configured quiet period.
quiet_hours_end	INTEGER	2	Ending hour of the configured quiet period.
is_active	BOOLEAN	1	Indicates whether the configuration is applicable.
created_at	DATETIME	-	Record creation timestamp.
updated_at	DATETIME	-	Record update timestamp.

Table 4.4.4 Database Table of Billing Configurations

4.4.5 Billing Statements Table

The billing_statements table stores billing statements data used by the implemented Smart Water application.

Field Name	Data Type	Length	Description
id	BIGINT	20	Primary key of the monthly statement.
user_id	BIGINT	20	Household billed by the statement.
billing_month	DATE	-	Month represented by the statement.
opening_balance	DECIMAL	12,2	Balance carried into the billing period.
total_liters	DECIMAL	18,6	Billable water consumption in liters.
total_cubic_meter	DECIMAL	18,6	Billable water consumption in cubic meters.
water_rate	DECIMAL	12,4	Rate snapshot applied at generation.
fixed_monthly_charge	DECIMAL	12,2	Fixed-charge snapshot applied at generation.
subsidy_deduction	DECIMAL	12,2	Subsidy snapshot applied at generation.
currency	VARCHAR	3	Statement currency.
current_charge	DECIMAL	12,2	Charge calculated for the current period.
amount_paid	DECIMAL	12,2	Payment recorded against the statement.
remaining_balance	DECIMAL	12,2	Outstanding balance after payment.
status	VARCHAR	255	Payment state such as unpaid, partial, or paid.
generated_at	DATETIME	-	Date and time the statement was finalized.
paid_at	DATETIME	-	Date and time full payment was recorded.
created_at	DATETIME	-	Record creation timestamp.
updated_at	DATETIME	-	Record update timestamp.

Table 4.4.5 Database Table of Billing Statements

4.4.6 Notifications Table

The notifications table stores notifications data used by the implemented Smart Water application.

Field Name	Data Type	Length	Description
id	BIGINT	20	Primary key of the notification.
user_id	BIGINT	20	Household recipient; nullable for administrative or batch scope.
water_reading_id	BIGINT	20	Related water reading when the notice is telemetry-driven.
type	VARCHAR	255	Notification category.
title	VARCHAR	255	Short notification heading.
message	TEXT	-	Full notification content.
metadata	JSON/TEXT	-	Structured context used by the interface or workflow.
is_read	BOOLEAN	1	Indicates whether the recipient opened the notice.
batch_id	VARCHAR	255	Identifier shared by a batch announcement.
delivery_status	VARCHAR	255	Delivery lifecycle state.
delivery_error	TEXT	-	Latest delivery failure message when applicable.
scheduled_at	DATETIME	-	Scheduled delivery timestamp.
sent_at	DATETIME	-	Actual delivery or creation timestamp.
archived_at	DATETIME	-	Administrative archive timestamp.
user_deleted_at	DATETIME	-	Household-side removal timestamp without source deletion.
created_at	DATETIME	-	Record creation timestamp.
updated_at	DATETIME	-	Record update timestamp.

Table 4.4.6 Database Table of Notifications

4.4.7 Leak Alerts Table

The leak_alerts table stores leak alerts data used by the implemented Smart Water application.

Field Name	Data Type	Length	Description
id	BIGINT	20	Primary key of the leak incident.
user_id	BIGINT	20	Affected household.
first_water_reading_id	BIGINT	20	First reading associated with the incident.
latest_water_reading_id	BIGINT	20	Most recent supporting reading.
notification_id	BIGINT	20	Notification associated with the incident.
status	VARCHAR	255	Incident state such as detected, acknowledged, resolved, or false alarm.
notes	TEXT	-	Administrative investigation and resolution notes.
detected_at	DATETIME	-	Initial detection timestamp.
last_detected_at	DATETIME	-	Most recent detection timestamp.
notified_at	DATETIME	-	Household notification timestamp.
acknowledged_at	DATETIME	-	Administrative acknowledgement timestamp.
resolved_at	DATETIME	-	Resolution timestamp.
created_at	DATETIME	-	Record creation timestamp.
updated_at	DATETIME	-	Record update timestamp.

Table 4.4.7 Database Table of Leak Alerts

4.4.8 Audit Logs Table

The audit_logs table stores audit logs data used by the implemented Smart Water application.

Field Name	Data Type	Length	Description
id	BIGINT	20	Primary key of the audit record.
user_id	BIGINT	20	Authenticated actor when available.
action	VARCHAR	255	Machine-readable administrative or security action.
auditable_type	VARCHAR	255	Type of affected domain record.
auditable_id	BIGINT	20	Identifier of the affected domain record.
description	VARCHAR	255	Human-readable activity description.
metadata	JSON/TEXT	-	Redacted structured activity context.
outcome	VARCHAR	255	Result such as success or failure.
ip_address	VARCHAR	45	Request IP address when retained.
user_agent	TEXT	-	Request client or browser description.
request_id	VARCHAR	255	Request correlation identifier.
created_at	DATETIME	-	Record creation timestamp.
updated_at	DATETIME	-	Record update timestamp.

Table 4.4.8 Database Table of Audit Logs

4.5 Development and Testing

Development is organized around feature-specific API controllers and domain services. WaterConsumptionService and HouseholdConsumptionService handle aggregation and availability semantics; BillingStatementService preserves monthly charge logic; AnalyticsService produces range-based summaries; and the embedded firmware handles acquisition, display, activation, and delivery reliability. Verification combines automated feature tests with a production frontend build.

Verification	Coverage / Output	Status	Duration	Observation
Laravel automated test suite	138 tests; 885 assertions	137 passed; 1 failed	18.017 seconds	The failing case expected one record but received two in an administrator consumption-history response.
Frontend production build	Vite 8.0.16; 4 modules transformed	Passed	6.75 seconds	CSS, boot JavaScript, main JavaScript, and manifest were produced successfully.

Current verification status: The project is substantially covered by automated tests, but the suite is not fully green as of 24 August 2026. One consumption-monitoring test must be investigated before claiming complete regression-test acceptance.

4.5.1 Major Test Areas

- Authentication and account restrictions, including blocked, suspended, archived, deleted, and unauthorized users.
- Municipal device registration, activation, hardware binding, credential rotation, reassignment, health, and decommissioning.
- Telemetry validation, device ownership, idempotent reading identifiers, server-authoritative totals, and precision.
- Household live readings, time-zone-aware periods, cumulative-reset handling, missing-data semantics, and billing estimates.
- Water-rate configuration, fixed monthly charges, statement generation, status changes, and historical billing fields.
- Leak alert creation, notification, resolution, false-alarm handling, reopening, and role protection.
- Administrator reporting, analytics, customer management, exports, notification lifecycle, and interface accessibility contracts.

4.6 Implementation Plan

Phase	Owner	Activity	Expected Output
1. Environment preparation	Municipal IT / Developers	Configure PHP, database, application key, mail/queue behavior, timezone, LAN URL, and production assets.	Environment checklist and successful health endpoint.
2. Database deployment	Developers	Back up the target database, run migrations and controlled seeders, and verify indexes and foreign keys.	Current schema and verified baseline records.
3. Administrator setup	System administrator	Create authorized staff accounts, set water rates and alert rules, and verify audit logging.	Operational administrative workspace.

Phase	Owner	Activity	Expected Output
4. Household onboarding	Authorized staff	Create household records with required address, contact details, role, status, and account number.	Active household accounts ready for assignment.
5. Meter commissioning	Technician + Administrator	Register each meter, record asset details, assign household, activate once, and verify last-seen health.	Bound device and successful telemetry.
6. Pilot verification	Project team	Compare measured volume with a calibrated reference, verify network recovery, review alerts, and reconcile billing estimates.	Signed pilot acceptance record.
7. Production rollout	Municipality	Deploy in phases, train staff and households, monitor audit/leak/device dashboards, and maintain backups.	Controlled operational adoption.

4.6.1 Implementation Issues and Challenges

Issue	Effect	Mitigation
Sensor calibration	Pulse-to-liter conversion varies by sensor, pipe conditions, pressure, and installation.	Perform volumetric calibration per installed meter and retain calibration records.
Network interruption	LAN or Wi-Fi loss can delay near-real-time visibility.	Use the firmware pending-reading state, stable IDs, health windows, and field connectivity checks.
Duplicate or noisy pulses	Electrical noise may inflate flow readings.	Use minimum pulse intervals, abnormal-burst rejection, and site validation.
Credential handling	Activation codes or tokens may be exposed during commissioning.	Issue short-lived single-use codes, remove codes after activation, rotate credentials, and restrict administrator access.
Billing configuration	Incorrect effective dates or rate settings can affect estimates and statements.	Require authorized review, retain charge snapshots, and reconcile pilot statements before rollout.
Test-suite regression	One current consumption-history test has a record-count mismatch.	Isolate the query/date fixture, correct the root cause, and rerun the complete suite before acceptance.

4.7 System Implementation and Results

The implemented application provides a complete path from a registered physical meter to role-scoped monitoring and administrative action. The meter reports measured intervals; the API validates device identity; the database preserves raw and derived values; services calculate consumption and estimated billing; and the user interfaces expose monitoring, leak, notification, billing, analytics, and reporting functions. The successful production asset build and 137 passing automated tests demonstrate broad functional coverage, while the remaining failing test identifies a specific acceptance item rather than an undocumented weakness.

Interpretation: The implementation is suitable for continued pilot testing and controlled demonstration. Production acceptance should follow calibration, backup, security, network-recovery, and full regression-test sign-off.